Audit, step 3, area C: full-wave capacitor co-simulation

Auditor: an independent agent (model: Fable), read-only, with synthetic numeric checks; no simulations run. HEAD 2734546. Fix status is in `DECISIONS.md` ("Audit actions").

ID	Severity	Status	Where	Finding
C1	Medium	CONFIRMED	`sim/cosim.py`	The quasi-static map is evaluated at the nominal plane position, but openEMS's H dump sits half a cell away (0.5 mm here, up to 1.5 mm in the phantom grid): a few percent of systematic bias in V18
C2	Medium (provenance)	CONFIRMED	DECISIONS	The official V15 run repeats the tagged thin-curve run (−10.3%), so V15 will FAIL; that was not recorded beforehand
C3	Medium (check design)	LIKELY	V18	The maximum is taken over every near-surface point, including the far side of the head where both maps are about 1% of their peak, so noise dominates. A registration weakness, not a code bug
C4	Low	CONFIRMED	Design §6b	Still says `AddWire`
C5	Low	LIKELY	`sim/run_queue.sh`	If an openEMS run fails, `cosim.py` still runs on a stale earlier `openems.npz`
C6	Low	POSSIBLE	`sim/cosim.py`	The root search has no residual check (a pole could be accepted); the reference value is the previous pass's root, not the first-pass value as documented
C7	Low	POSSIBLE	`cosim.py` L extraction	Gap and port parasitic capacitance is absorbed into L_raw and hence into a_eff; the gap fraction differs between the loop and the trefoil
C8	Low	POSSIBLE	export vs app	V19's d_o uses `zeros[last]`, while the app uses `zeros[0]` (identical here)
C9	Low	POSSIBLE	`cosim.py`	`inside` uses the analytic ellipsoid, while the phantom is 3 mm voxels; depth is measured along the ray, not as the nearest surface distance

Verified correct

- **Network maths on a synthetic loop:**
  - capacitor values reproduced to 12 digits;
  - S11 −50.7 dB;
  - the preamp-decoupled load and the 1 W source convention behave as expected;
  - the wave round trip closes to 1e-15.

- **Sign and orientation conventions:** intra-element results don't depend on port orientation; the M_eff sign check is meaningful.
- **Thevenin source.**
- `cut_wire` on all exported cases.
- **Frequency handling:** f0 falls exactly on a sample.
- **M3 superposition-check logic.**
- **Thin-wire correction:** the trefoil's correction is 100.5 nH against a 25 nH tolerance, so V15b has real power as a transfer test.
- **Units, $B1^{-}$ convention, phantom writer.**
- **Verdict logic in `tests/run.js`.**